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Solar PV array-based DC–DC converter with MPPT for low power applications

Akhil Raj, Sabha Raj Arya* and Jyoti Gupta

Department of Electrical Engineering, Sardar Vallabhbhai National Institute of Technology, Surat 395007, India

This article discusses a DC–DC converter based solar fed PV array system for low power applications. A single diode based solar panel is designed and modelled for this PV array system. At low value of irradiance in solar panel, the system will perform with better accuracy. To acquire better efficiency, the similar system is operated with Maximum Power Point Tracking (MPPT) control algorithm. The solar panel current v/s Voltage (I–V) and Power v/s Voltage (P–V) characteristics curves are obtained from Matlab simulation. The comparison of simulation curves is also done with the curves obtained from the experiment results. The detailed investigation of DC–DC converters with solar fed system with soft-switching technique is also performed. This soft-switching technique is formed by using the resonant inductors and capacitors. These passive components reduce the circuit switching losses. The Zero Voltage Switching (ZVS) technique is used to eliminate the switching losses developed in the circuit at the time of switching transitions. This technique also reduces the dv/dt noise owing to junction capacity discharge. This Proposed ZETA and SEPIC converters based system are modelled and simulated in MATLAB program and its performance is evaluated through experimentation in the terms of losses and efficiency. From the performance study of ZVS ZETA and SEPIC converter, increment in efficiency are obtained 1.97% and 7.23% under Perturb and Observe MPPT control and 4.14% and 0.41% in Incremental Conductance MPPT control respectively as compared to its conventional ZETA and SEPIC converter.

Introduction

In comparison with conventional energy sources, renewable energy sources can provide clean energy with optimal and effective utilization [1]. Compared to other renewable energy resources, solar-based power generation is readily accessible [2]. It is widely used for charging the battery, lighting and solar pumping etc. Energy problem is minimized with the usage of pollution-free renewable energy without emissions and with the maximum power output using solar energy with smart grid system through optimization [3,4]. Reddy and Bijwe [5] have discussed about the economics involve in real time load dispatch considering system based on renewable energy. The primary issues in the solar-based scheme are low efficiency and energy loss. Antonio *et al.* [6] described about the conventional power monitoring technique in the solar PV system is design to operate under uniform environmental conditions and why it is fail under non-uniform condition. For extracting maximum power with less oscillations, the microcontroller-based control technique is described in the literature [7]. Because of considering the variation in solar irradiation, voltage and current, it is very difficult to design a solar PV module for experimental uses. The Detail of mathematical modelling of solar panel is illustrated in the literature [8]. and the mathematical study of the photovoltaic module is described in Ref. [9]. A Solar PV array application can be categorized into two classes is home and business applications. The Design of a solar-powered battery

*Corresponding author. Raj, A. (akhilraj0110@gmail.com), Arya, S.R. (sabharaj79@gmail.com),

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charging implementation is elucidated in Refs. [10,11]. The Various energy monitoring efficient techniques are used to achieve for maximum power such as Perturb and Observation MPPT technique, Incremental Conductance MPPT method and Constant Voltage MPPT method [12]. Amid these methods, the best cost effective is Perturb and Observation (P&O) method [13]. The safety is taken into the consideration at the time of formulation of solar panel used in some applications with minimal downtime are discussed in the literature [14].

The solar panel's developed dc voltage can be converted into the required logical voltage levels. This voltage transformation tends to occur through dc–dc converters and storage though battery based system [15,16]. The core operational concepts and method for conversion was deliberated in Ref. [17]. Tymerski and Vorperian [18] have discussed various classifications of PWM DC–DC converter in terms of converter cell and its small signal analysis is done. The paper [19] provides a comprehensive explanation on the ground network theory of the distinct characteristics of the converters. Nowadays, two inductor converter topology, high step down converter used other than ordinary dc–dc converters discussed in Ref. [20,21]. Shi *et al.* [22] has discussed the effect of parasitic elements in tapped inductor based dc/dc converters.

The effectiveness of a regular converter varies from 85 to 90%. This can be enhanced by implementing the idea of soft switching to 90–95% [23]. The hard switching approach used in converter, the switching loss occur across switch is very high. However, the soft-switching approach used in converter will decrease the losses at the time of switching. The soft switching strategy is accomplished by using resonant inductors and capacitors and diode [24–26]. Amongst Zero Voltage Switching (ZVS) and Zero Current Switching (ZCS) soft-switching method, the more precise method is ZVS. In ZCS method, due to diode peak inversion current in converter circuitry is high that will harm the circuit design. Hence, the ZVS strategy is widely used [27]. In Refs. [28] and [29], the ZVS and ZCS comparative analysis is provided in depth. This comparative analysis depends on type of performance and non-isolated converter features such as boost, buck, Cuk, single-ended primary-inductor converter (SEPIC) and ZETA is described in the literature [30].

This article discusses and investigates the DC–DC ZETA and SEPIC converters use in solar PV array application. This solar system performance is also analysed for zero voltage switching (ZVS) based ZETA and SEPIC converter. The drawback occurs in ZETA converter are ripple present at load side is more than SEPIC converter because inductor at output side of SEPIC converter smooth the current ripple. Moreover, the main benefits of these converters are Non-inverted outputs with step-down or step-up. These input voltage and current is supplied by solar panel. The system performance is noted conventional switching and under ZVS circumstances. The MPPT monitoring strategy for switching action is used to obtain maximum energy from the solar panel. The converters performance analysis is done by using MPPT methods with solar Perturb and Observation (P & O) and Incremental Conductance (IC).

Conventional and resonant converters

The conventional and resonant converters basic differences are switching techniques. Two kinds of switching techniques are presents hard and soft switching. In hard switching, the transition in switching does not depend on voltage and current across switch whether the voltage and current across switch is zero or not. The gate pulses energize the switch randomly at any instant. Therefore, the switching losses across switch, voltage and current stress across switch and EMI are high. The switch voltage will reduce and switch current will increase at switch turn-on moment. Therefore, the energy loss will happen for a tiny period during the turn-on moment of voltage and current across switch. It is recognized as switch energy loss. For smooth switching operation, the device turns on after the device voltage becomes zero. Thus, either voltage or current across switching device becomes zero during switching, that will become switch energy loss to zero. This technique is termed as ZVS. Similarly, the switch turns on when the current passes through the switch becomes zero known as ZCS technique [17].

The ZVS switching method is designed to reduce recovery losses across the diode and the output capacity of semiconductor caused by switching losses in the MOSFET switch. By using the ZCS switching method for the same switch, there is no impact of diode reverse recovery losses as well as MOSFET output capacitance resulting from switching loss. In corollary, there is no sign of improving the converters efficiency in ZCS switching technique where MOSFET is use as switching device [23]. In this section, two resonant converters configurations are discussed.

Solar fed ZVS based resonant ZETA converter [15,17,30]

The ZETA converter is obtained from DC–DC buck boost converter. It can generate input voltage either less or more than output voltage. This topology offers reduced ripple output voltage and easier compensation compared to DC–DC SEPIC converter topology. The other benefits are the constant (non-pulsating) output and dc isolation between input and output. Figure 1 (a) depicts a DC–DC conventional ZETA converter topology. It comprises of inductors L1, L2 (here, L2 serve as filter inductor), intermediate capacitor C1 which helps to transmit energy from input to output side through output capacitor C2. When switch turns ON, the diode D will act as reverse biased and the input and output inductors L1, L2 starts charging with the same polarity depicts in Figure 1(a). The voltage across both the inductors is equal to supply voltage Vs. when the switch turns off and the diode D is in ON condition, and then the voltage through the inductors L1 and L2 is negative ($-V_o$).

The Proposed resonant ZVS based ZETA topology with solar power supply is depicted in Figure 1(b) where switch is ON and OFF at zero voltage. Figure 1(b) shows the ZVS based ZETA Converter using the MPPT control technique for switch transition ON/OFF state. In this circuit, the ZVS operation is achieved by using inductor (L_r), capacitor (C_r) and the diode is connected parallel across the MOSFET switch. The switch's inner junction condenser is added to the resonant condenser that will only influence the resonance frequency but not the switching frequency. During the switching transitions that inner junction condenser adds energy loss throughout the switch. Therefore, If the MOSFET switch is use, then the capacitor voltage (Cr) is blocked by the diode. That will improve converter performance.

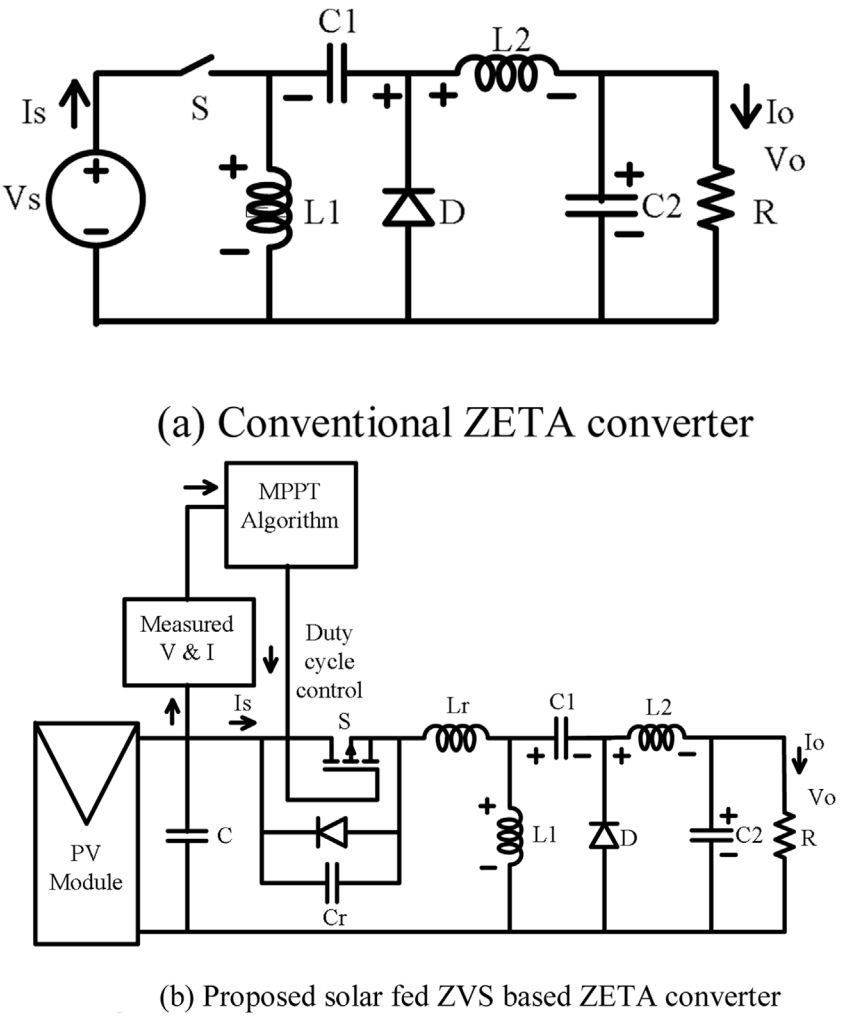


FIGURE 1

(a) Conventional ZETA converter. (b) Proposed solar fed ZVS based ZETA converter.

Solar fed ZVS BASED resonant SEPIC converter [15,17,30]

The SEPIC converter generates a synchronized voltage of the favourable output. It can produce more or less output voltage than the input voltage. The SEPIC converters are used in which applications where the voltage across battery should be less or more than the required voltage level. This circuit comprises input and intermediate inductors L1, L2 and intermediate and output capacitors C1, C2 shown by Figure 2(a)

For continuous conduction mode (CCM) of operation of SEPIC converter, ripple voltage across intermediate capacitor C1 is neglected. Therefore, the C1 coupling capacitor charges the voltage V_s continuously. When the switch gets off, the voltage across input inductor V_{L1} is equal to the V_{out} and voltage across switch is sum of input and output voltages. The current is flowing through L1, C1, D and into C2 at the time of switch OFF condition. In OFF period, capacitors C1 and C2 get re-energized and provide load current to charge the input inductor L1. When the switch (S) turns ON, the intermediate capacitor (C1) is in parallel with intermediate inductor L2, the intermediate inductor voltage V_{L2} will be equal and opposite to intermediate capacitor voltage V_{C1}. When switch-ON period, the energy stored in input inductor L1 from the input supply and intermediate inductor L2 stored energy from releasing energy of intermediate capacitor C1. The proposed Solar fed ZVS based SEPIC converter is depicted in Figure 2(b)

The ZVS operation is accomplished by using capacitor and diode connected in parallel with the switch and series inductor connection with the switch. Here, the used ZVS switching technique is M-type switching technique. This converter's procedure will be the same as previously stated.

Design of solar fed ZETA and SEPIC converters

Solar cells are introduced to convert solar power into electrical power. It consists of metals with semiconductors and does not require high temperature to convert the energy. The energy emitted from the sun is in the form of ultraviolet, visible waves or infrared radiations. When the sun shines and touches the surface of the solar cells, these waves migrate towards the earth. One solar cell generates several quantities of electricity, which may be enough for high-power applications. Several solar cells are connected in the formation of solar panel. The typical use of solar cell quantity in the solar panel is mostly 32. Several solar panels are needed to generate enough electricity to provide electricity to consumers at home. The power converter design used in solar fed system is described below:

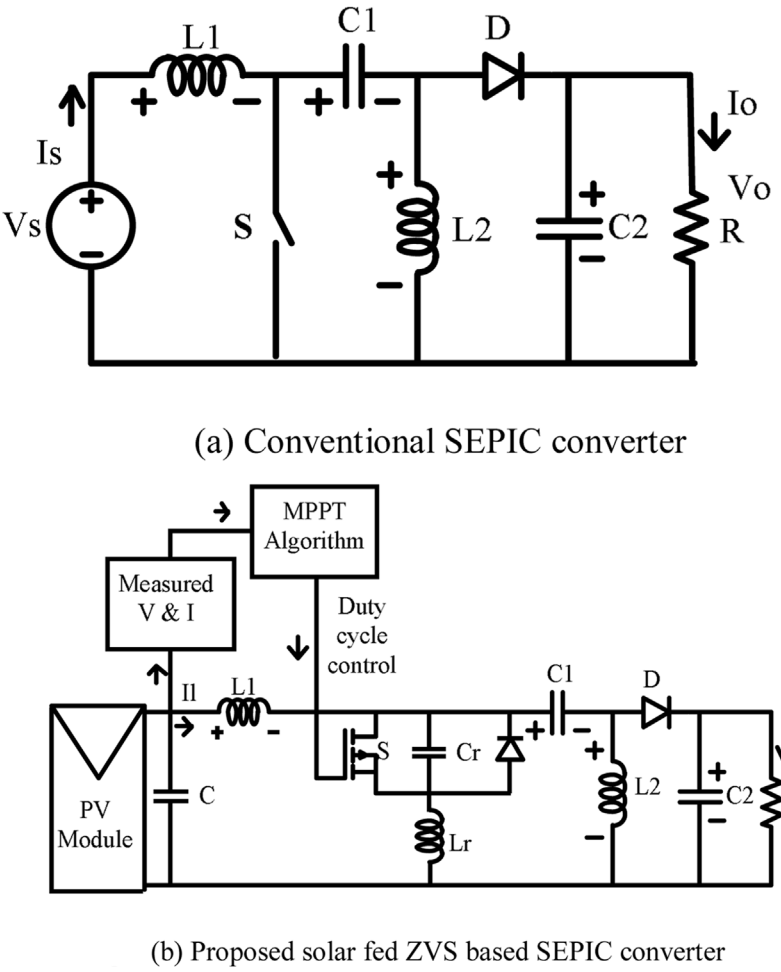


FIGURE 2

(a) Conventional SEPIC converter. (b) Proposed solar fed ZVS based SEPIC converter.

Design of solar fed ZETA converter [17,21]

The Continuous Conduction operating Mode (CCM) of ZETA converter for the duty cycle (D) can be defined as a function of output voltage (V_{out}) and voltage input (V_{in}). The expression of duty cycle (D) is given as,

$$D = \frac{V_{out}}{V_{out} + V_{in}} \tag{1}$$

The important parameter should be considered in the design of DC–DC ZETA converter is inductor ripple current (ΔI_L). Therefore, the peak-to-peak value of required ripple currents,

$$(\Delta I_{L(pp)}) = KI_{in} = KI_{out}\frac{D}{1-D} \tag{2}$$

here, K = ripple current constant (0.2 to 0.4).

The design equation of minimum value of inductors L_1 and L_2 shown in (3) by using (1) and (2),

$$L_{1(\min)} = L_{2(\min)} = \frac{1}{2}\frac{V_{in}D}{\Delta I_{L(pp)} f_s} \tag{3}$$

The peak value of current flow through inductor L_1 and L_2 can be written as a function of duty ratio and the ripple current is given in (4),

$$I_{L_1(pk)} = I_{out}\frac{D}{1-D} + \frac{\Delta I_L}{2} \text{ and } I_{L_2(pk)} = I_{out} + \frac{\Delta I_L}{2} \tag{4}$$

The design equation of minimum value of the output capacitance or the filtering capacitance (C_2) is given as,

$$C_{2(\min)} = \frac{\Delta I_{L(pp)(actual)}}{8\Delta V_{c(out)} f_s} \tag{5}$$

Here, $\Delta V_{c(out)}$ = Output ripple voltage (typically 1%–2% of V_{out})

The components values taken in ZETA converter design are specified in Appendix-A.

Design of solar fed SEPIC converter [17,21]

To find out the maximum switching current of circuit, it is necessary to find out the expression for the duty ratio. The duty ratio is nothing but a fraction of one period in which the signal is active. Therefore, the expression for the duty ratio (D) of a SEPIC converter is given by,

$$D = \frac{V_{out} + V_{fwd}}{V_{in} + V_{out} + V_{fwd}} \tag{6}$$

Where, V_{fwd} = Drop in forward voltage of the Schottky diode

The above expression (6) is rearranged and can be re-written as:

$$\frac{D}{1-D} = \frac{V_{out} + V_{fwd}}{V_{in}} = \frac{I_{in}}{I_{out}} \tag{7}$$

Duty ratio value comes from (7) maximum and minimum at minimum and maximum input voltage respectively. Form above Eq. (7), it is observed that the duty ratio can be represented in terms of input current (I_{in}) and output current (I_{out}). The ripple in inductor current (ΔI_L) is a function of efficiency (η) can be calculated as,

$$\Delta I_L = (20 \text{ to } 30\%)\frac{I_{in}}{\eta} \tag{8}$$

For the required value of the ripple current, the minimum inductor value can be calculated as,

$$L_{1(\min)} = L_{2(\min)} = \frac{1V_{in(\min)}D_{\max}}{2 \quad \Delta I_L f_{sw}} \tag{9}$$

The peak current flowing in inductors is calculated by Eq. (10),

$$I_{L1(pk)} = \frac{I_{in}}{\eta} + \frac{\Delta I_L}{2} \text{ and } I_{L2(pk)} = I_{out} + \frac{\Delta I_L}{2} \tag{10}$$

The expression for Coupling capacitor C_1 and filter capacitor C_2 is given as,

$$C_1 = \frac{I_{out}D_{\max}}{\Delta V_{C1} f_{sw}} \tag{11}$$

Where, ΔV_{c1} = Ripple voltage in the coupling capacitor C_1 (1% ripple of input voltage V_{in})

$$C_2 \geq \frac{I_{out}D_{\max}}{\Delta V_{Cout} f_{sw}} \tag{12}$$

Where, ΔV_{Cout} = Output ripple voltage (typically 1% of V_{Cout})

The design component value of SEPIC converters are specified in Appendix-A.

Design of resonant parameters [23]

The switching and resonance frequencies are the two major things during the design of L and C in a circuit. The component size is high when the value of switching frequency is less. The resonant frequency decides the mode of the operating region. If the switching frequency is less than the resonance frequency, then the operation will work under ZVS mode and if it is less, the it will work under ZCS mode. At the resonant condition, the inductive and capacitive reactance is considered as equal. Both the Zeta and SEPIC converters discussed in this paper is a derived as buck-boost converter. The quality factor (Q) of the buck-boost circuit can be fixed from the graph between voltage gain (M) and normalised switching frequency (f_{ns}). The equations used for the calculation of resonant circuit parameters are given below.

$$\text{The switching frequency, } (f_s) = \frac{1}{T_s} \tag{13}$$

$$\text{The normalised frequency, } f_{ns} = \frac{f_s}{f_o} \tag{14}$$

The load impedance (Z_o) of the converter as a function of load resistor (R_o) and quality factor (Q) can be defined as,

$$Z_o = \frac{R_o}{Q} = \sqrt{\frac{L_r}{C_r}} \tag{15}$$

$$\text{Therefore the resonant frequency, } (f_r) = \frac{1}{2\pi\sqrt{L_r C_r}} \tag{16}$$

The voltage gain (M) is the ratio of output to the input voltage and it is given as,

$$M = \frac{V_o}{V_{in}} \tag{17}$$

where L_r = Resonant inductor (Henry); C_r = Resonant capacitor (Farad).

Control scheme

The MPPT control scheme monitors the transition used in solar fed DC-DC ZETA and SEPIC converters. At one instant of operating point exits in photovoltaic modules gives the maximum energy extraction. But the PV module operating point is essential to monitor for maximum energy extraction is done by MPPT control technique. This method will always attempt to keep the panel's operating point at the maximum power operating point. Gate pulses across switch produced by the MPPT method is compared to the converter's switching frequency and generate the duty ratio related to the maximum Power Point. Amongst various MPPT techniques, only the perturbation and observation and incremental conductance techniques are described as:

Perturb and observe (P and O) method [12]

The P & O MPPT method will disturb and observe the energy along with the solar module's energy curve. It begins from a certain stage on Power–Voltage (P–V) curve then set the value of original duty cycle for estimation of this stage energy and provides the beneficial disturbance according to the curve's next point. The system is started at (x-1) value is equal to zero. The next perturbation is still follows identical path as mentioned above and it also measures the solar panel voltage V_{pv}. Then, the solar system will verify whether the ΔP value is more than or equal to 0. If the ΔP value is more than 0 then it should be tested in same direction as perturbation direction. If ΔP value is more than 0 gives the increasing module voltage V_{PV} and if it is not, the PV module voltage will be replaced with certain instant past values. If the ΔP is less than 0, the whole cycle would be interrupted in the opposite direction as described above.

Incremental conductance (IC) method [12]

The IC MPPT technique is designed to overcome the limitation of P & O MPPT techniques. The Incremental and instantaneous conductance of solar PV module are used to monitor the maximum point at which the maximum energy can be drawn is given below as,

$$\frac{I}{V} = \frac{dI}{dV} \tag{24}$$

where,

$\frac{I}{V}$ = Instantaneous conductance

$\frac{dI}{dV}$ = Incremental conductance

When the power shift to voltage level ratio is zero, the full control point may be accomplished. So according to derivative rule,

$$\frac{dP}{dV} = \frac{d(V*I)}{dV} = I\frac{dV}{dV} + V\frac{dI}{dV} \tag{25}$$

At max power point, $\frac{dP}{dV} = 0$ therefore, the incremental and instantaneous conductance becomes equal and opposite given in (26).

$$\frac{dI}{dV} = -\frac{I}{V} \tag{26}$$

If $\frac{dP}{dV} > 0$, This implies the PV voltage is lower than the MPP voltage, for this case the duty cycle should be increased. If $\frac{dP}{dV} = 0$ (PV voltage is equal to the MPP) and $\frac{dP}{dV} < 0$ (PV voltage is more than the MPP) in both the cases, duty cycle should be reduced.

The real peak energy from the solar panel is found by P & O and IC MPPT control techniques. The IC technique provides a precise value as compare with P&O technique. Here, with P & O technique oscillations occur near to the peak value when the atmospheric circumstances differ slowly. In addition, the IC Method offers the exact value of atmospheric change.

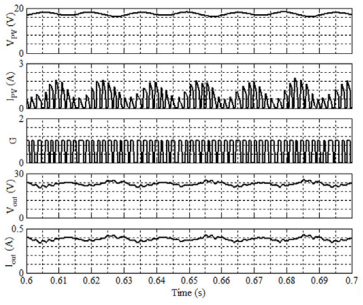
Simulation results

The solar fed ZETA and SEPIC converter based PV array system is simulated in MATLAB software under constant value of solar irradiance 1000 W/m^2 and temperature 25 °C. The system is also analysed for dynamic performance under variation in solar irradiance. The switching frequency used for DC–DC ZETA and SEPIC converter is 5 kHz. By using the resonant parameters in ZVS technology, switching losses in the circuit reduces that will enhance the converter characteristics. The simulation results are taken Under P&O and IC MPPT control algorithm in solar PV array system based on ZETA and SEPIC converter technology. And these performance results are compared with ZVS based converter system. It is noted that the IC MPPT control technique is more effective than the P&O control technique proved by simulation results shown by Figures 3–5. The simulation data used for ZETA and SEPIC converters are specified in Appendix.

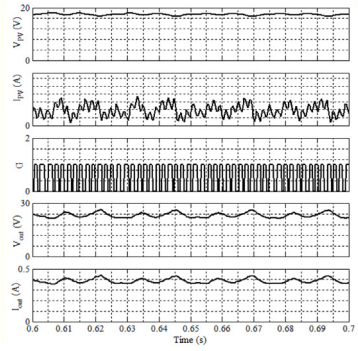
Performance study of solar-fed conventional and ZVS based ZETA and SEPIC converter with perturb and observation (P and O) MPPT control technique

Figure 3(a–d) depicts conventional and ZVS based ZETA and SEPIC converter output results by using P and O MPPT control technique. Figure 3(a, b) shows output waveforms of voltage (V_{PV}), current (I_{PV}), gate pulse (G) obtained from solar PV panel and obtained output of DC–DC converter are voltage (V_{out}), current (I_{out}), resonant capacitor's voltage (V_{Cr}) and resonant inductor current (I_{Lr}). From this analysis, it is observed that the voltage of photovoltaic device is non-linear in nature. The input current (I_{PV}) of ZETA converter is discontinuous but the input current of SEPIC converter is constant in nature which is depicted in Figure 3(a, b). The gate pulse of switch is produced by MPPT control technique. The P & O control execution is done after the estimation of solar panel voltage (V_{PV}) and current (I_{PV}). From Figure 3 (a, b), it is observed that ripple in output voltage of ZETA converter is less than ripple in output voltage of SEPIC converter. In ZETA converter, using series inductance at output side prevents the sudden changes occur across output current that will rise ripple across the output voltage. As compare to SEPIC Converter input voltage ripple of ZETA converter is high.

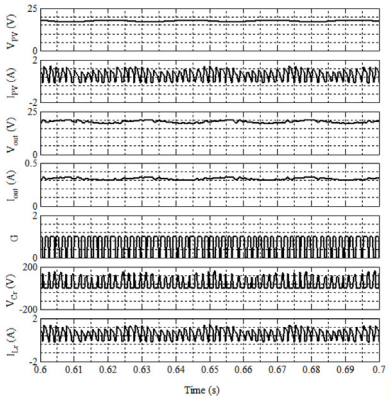
ZVS method used in ZETA and SEPIC converters is to make reduction in switching losses. Figure 3(c, d) shows switch is turn



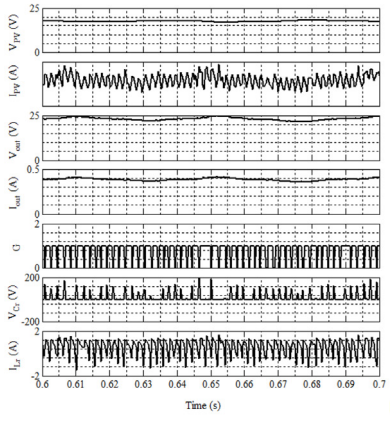
(a) Simulation results of solar fed conventional ZETA converter with P & O MPPT control technique



(b) Simulation results of solar fed conventional SEPIC converter with P & O MPPT control technique



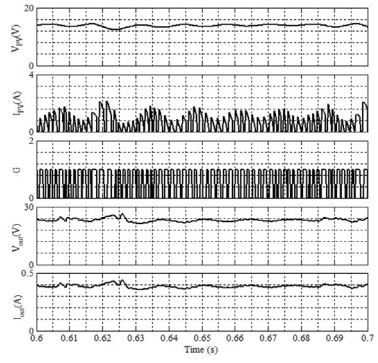
(c) Simulation results of solar-fed ZVS based ZETA converter with P & O MPPT control technique



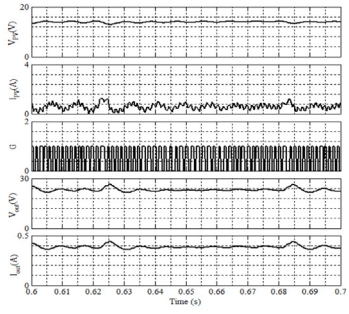
(d) Simulation results of solar-fed ZVS based SEPIC converter with P & O MPPT control technique

FIGURE 3

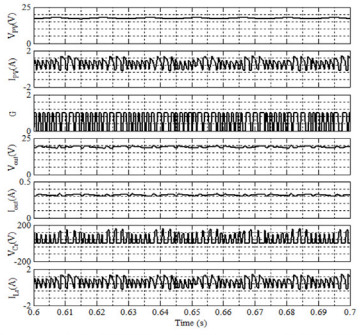
(a) Simulation results of solar fed conventional ZETA converter with P & O MPPT control technique. (b) Simulation results of solar fed conventional SEPIC converter with P & O MPPT control technique. (c) Simulation results of solar-fed ZVS based ZETA converter with P & O MPPT control technique. (d) Simulation results of solar-fed ZVS based SEPIC converter with P & O MPPT control technique.



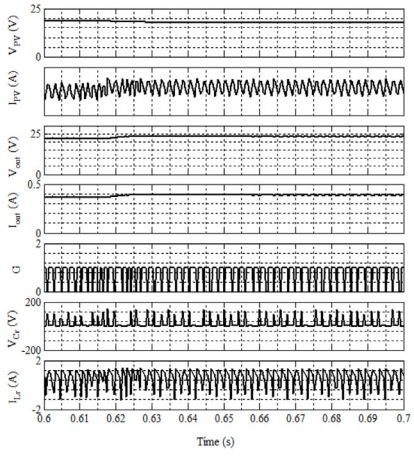
(a) Simulation results of solar fed conventional ZETA converter with IC MPPT control technique



(b) Simulation results of solar fed conventional SEPIC converter with IC MPPT control technique



(c) Simulation results of solar-fed ZVS based ZETA converter with IC MPPT control technique



(d) Simulation results of solar-fed ZVS based SEPIC converter with IC MPPT control technique

FIGURE 4

(a) Simulation results of solar fed conventional ZETA converter with IC MPPT control technique. (b) Simulation results of solar fed conventional SEPIC converter with IC MPPT control technique. (c) Simulation results of solar-fed ZVS based ZETA converter with IC MPPT control technique. (d) Simulation results of solar-fed ZVS based SEPIC converter with IC MPPT control technique.

on at zero voltage because of using ZVS method. Here, Figure 3(c, d) shows voltage across the resonant capacitor and current across resonant inductor are V_{Cr} and I_{Lr} respectively. Therefore, the switching stress across the switch is reduced and that will improve the efficiency of the converter. With this ZVS method, ZETA converter efficiency is enhanced approximately by 1.97% and efficiency of SEPIC converter is enhanced by 7.23%.

Performance study of solar fed conventional and ZVS based ZETA and SEPIC converters with incremental conductance (IC) MPPT control technique

Figure 4(a–d) shows conventional and ZVS based ZETA and SEPIC converter output analysis is done by using IC MPPT control technique. The figure shows waveforms of input voltage (V_{PV}), input current (I_{PV}), gate pulses (G), voltage across output (V_{out}), current across output (I_{out}), resonant condenser voltage (V_{Cr}) and resonant inductor current (I_{Lr}). The benefits of using IC MPPT control over P & O MPPT control are require less time to attain maximum power point. In addition, it provides more effectiveness as compared to P & O MPPT control technique. In IC MPPT control technique only drawback is high execution cost.

The ZVS based ZETA and SEPIC converters by using IC MPPT control technique results depicted in Figure 4(c, d). With this ZVS technology, The ZETA converter efficiency is enhanced by 4.14% and SEPIC converter efficiency enhanced by 0.41%. Therefore, ZETA converter is more efficient than SEPIC converter. The comparative study of conventional and ZVS based ZETA and SEPIC converters with P & O and IC MPPT control techniques are presented in Table 1.

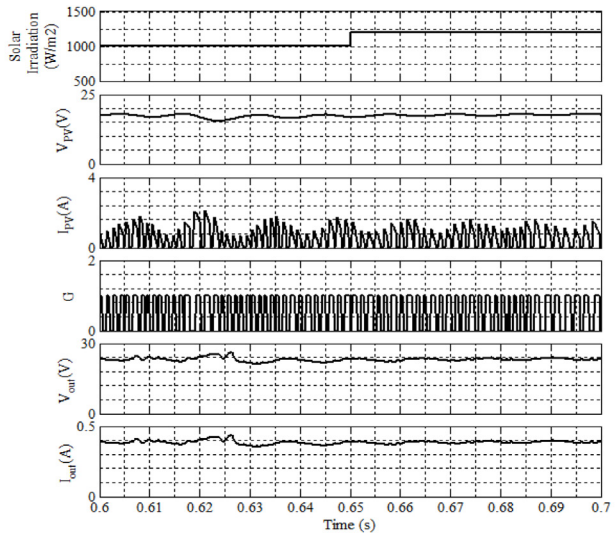
Table 1 shows the performance parameters achieved in ZVS based ZETA converter with IC MPPT method provides more effectiveness with high efficiency among all other configurations.

Dynamic performance of ZETA converter under variation in solar irradiance

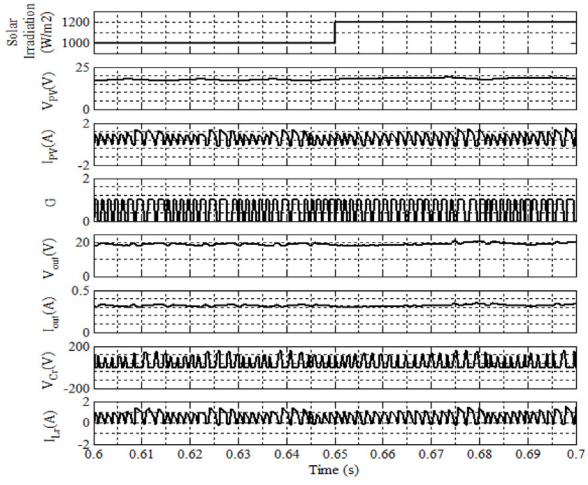
From the Table 1, It is concluded that the conventional and ZVS based ZETA converter with IC MPPT control technique has higher efficiency as compare to other configurations with IC and P & O MPPT control techniques. Therefore, the variation in solar irradiation is applied to ZETA converter with IC MPPT control technique shown by Figure 5 (a). The ZVS based ZETA converter topology performance under varying irradiation is shown by Figure 5(b). With the variation in solar irradiation from 1000 W/m^2 to 1200 W/m^2, the terminal voltage of the panel is enhanced that will increase the converter output power. Figure 5(a, b) shows irradiation value is 1000 W/m^2 up to 0.65 s and at 0.65 s it is changed to 1200 W/m^2 up to 1 s.

Test results

The DC-DC ZETA converter based solar PV array system is implemented for low power applications. The Incremental Conductance (IC) MPPT control based solar PV array model is implemented in Real-time simulator based on OP-5142 with 20 μs sampling time. Four channel Digital Storage Oscilloscope (DSO) is used for capture the response of system in form of waveforms. The switching



(a) Simulation results of ZETA converter with IC MPPT control technique under variation in solar irradiance



(b) Simulation results of ZVS ZETA converter with IC MPPT control technique under variation in solar irradiance

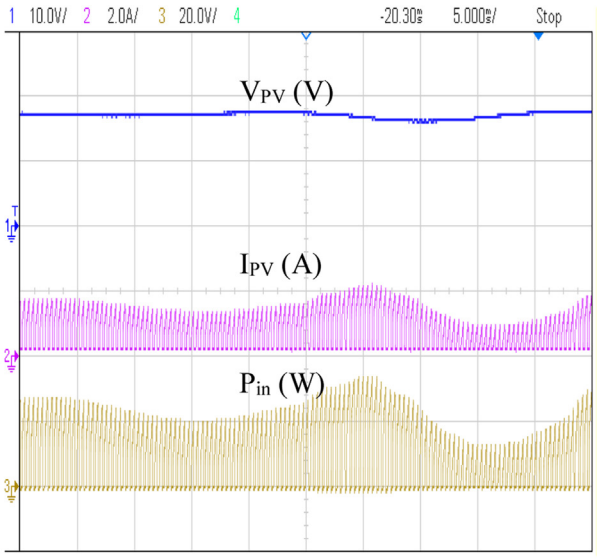
FIGURE 5

(a) Simulation results of ZETA converter with IC MPPT control technique under variation in solar irradiance. (b) Simulation results of ZVS ZETA converter with IC MPPT control technique under variation in solar irradiance.

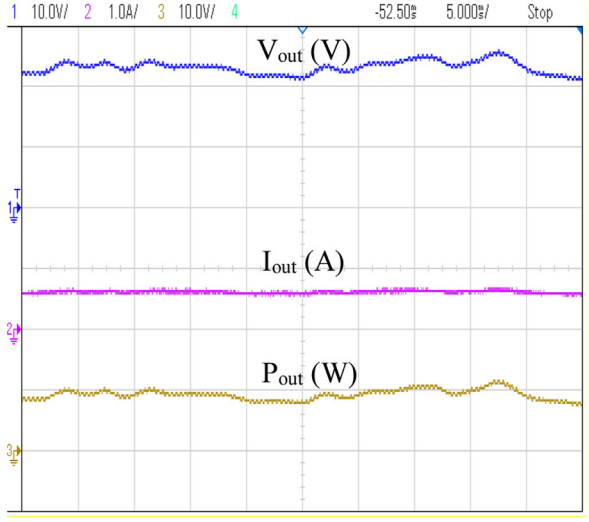
TABLE 1

Converter performance parameters at 1000 W/m^2 solar irradiance.

Converter	V_{in} (V)	I_{in} (A)	P_{in} (W)	V_{out} (V)	I_{out} (A)	P_{out} (W)	Losses (W)	Efficiency (%)
P & O MPPT control technique								
Zeta	17.26	0.58	10.01	22.95	0.3826	8.78	1.23	87.71
ZVS Zeta	17.26	0.58	10.01	23.21	0.3868	8.977	1.157	89.68
SEPIC	17.41	0.56	9.796	22.14	0.369	8.169	1.626	83.39
ZVS SEPIC	17.47	0.58	10.13	23.47	0.3912	9.181	0.95	90.62
IC MPPT control technique								
Zeta	17.00	0.5428	9.2276	22.81	0.3802	8.672	0.555	93.98
ZVS Zeta	17.00	0.5428	9.2276	23.31	0.3885	9.055	0.1726	98.12
SEPIC	17.23	0.5787	9.97	23.02	0.3837	8.832	1.1372	88.59
ZVS SEPIC	17.23	0.5787	9.97	23.06	0.3844	8.864	1.106	89.00



(a) Magnitude -axis: Ch1-10V/div, Ch2-2A/div, Ch3-20V/div; Time -axis: 5ms/div



(b) Magnitude -axis: Ch1-10V/div, Ch2-1A/div, Ch3-10V/div; Time -axis: 5ms/div

FIGURE 6

(a–b) Experimental results of solar fed conventional ZETA converter with IC MPPT control algorithm.
(a) Magnitude-axis: Ch1-10V/div, Ch2-2A/div, Ch3-20V/div; Time-axis: 5 ms/div.
(b) Magnitude-axis: Ch1-10V/div, Ch2-1A/div, Ch3-10V/div; Time-axis: 5 ms/div.

frequency used for DC–DC converter in the system implementation is chosen to be 5 kHz. The brief discussion of output waveforms of conventional and ZVS based ZETA converter is depicted in Figures 6 and 7 respectively.

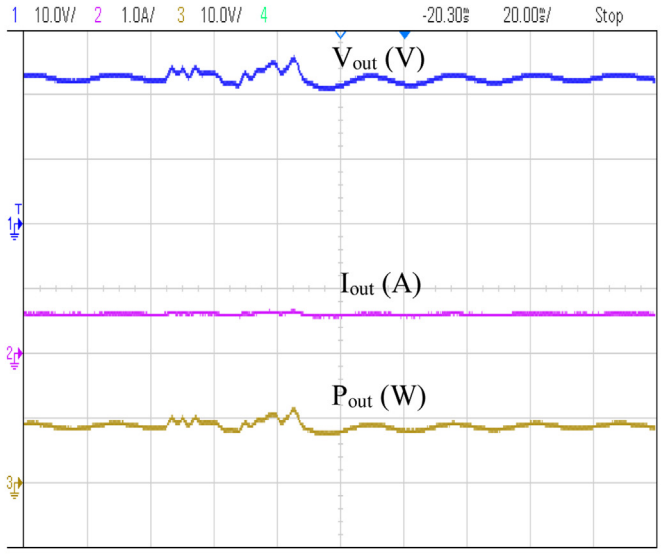
Solar-fed conventional ZETA converter performance with IC MPPT control algorithm

Figure 6(a) shows the input side performance of conventional ZETA Converter in terms of voltage (V_{PV}), current (I_{PV}) and power (P_{PV}). This figure shows average value obtained from the test results are 17 V, 0.5A and 9.2 W respectively same as results obtained from simulation. This experimental operation of solar panel is supplied equal voltage and current at maximum power point for full output requirement. Here, the experimental performance is verified for constant value of irradiance 1000 W/m^2 and temperature 25 °C.

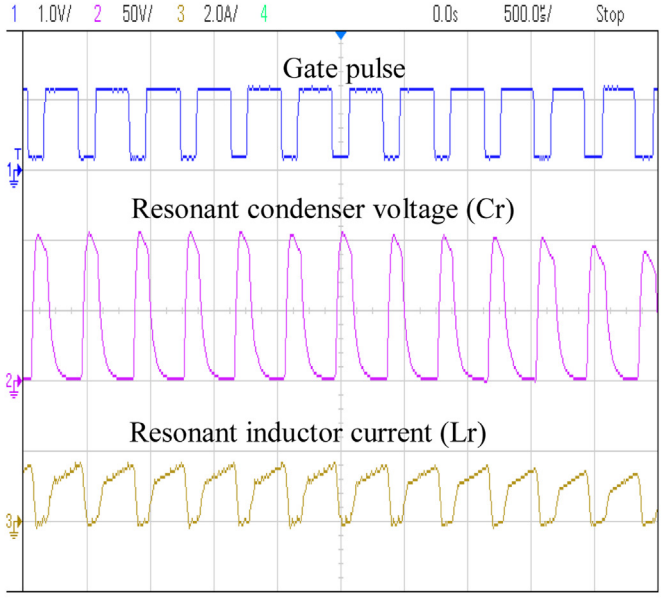
Figure 6(b) shows the output side performance of same converter in terms of voltage, current and power as Ch1, Ch2 and Ch3 respectively. The shown output side hardware experimental results are same as the output side simulation results. The obtained output average power from experiment is 8.6 W and the entire losses obtained in the circuit are 0.6 W. As a result, the conventional ZETA converter obtained efficiency is 93.47%.

Solar-fed ZVS-based ZETA converter performance with IC MPPT control algorithm

This solar fed converter system efficiency is analysed by using ZVS technologies. The advantage of ZVS technology used here is to



(a) Magnitude -axis: Ch1-10V/div, Ch2-1A/div, Ch3-10V/div; Time -axis: 20ms/div



(b) Magnitude-axis: Ch1-1V/div, Ch2-50V/div, Ch3-2A/div; Time-axis: 500µs/div

FIGURE 7

(a–b) Experimental results of solar fed ZVS based ZETA converter with IC MPPT control algorithm.
(a) Magnitude-axis: Ch1-10V/div, Ch2-1A/div, Ch3-10V/div; Time-axis: 20 ms/div.
(b) Magnitude-axis: Ch1-1V/div, Ch2-50V/div, Ch3-2A/div; Time-axis: 500 µs/div.

reduce the switching losses. The reference input quantities are assumed to be same as conventional converter. Figure 7 depict output voltage (V_{out}), current (I_{out}), power (P_{out}), gate pulses, resonant condenser voltage (V_{Cr}) and resonant inductor current (I_{Lr}) waveforms. By applying ZVS in traditional zeta converter, the converter's energy will improve. Figure 7 (a) shows output power is 9 W. The losses in solar fed ZVS based ZETA converter are 0.2 W lower than the conventional converter. Therefore, the ZVS based ZETA converter efficiency is improved to 97.8 % which is 4.3% more than the conventional converter. Figure 7(b) depicts the voltage across switch becomes zero whenever switch becomes turn on, that satisfied the ZVS condition.

Conclusion

This article presents a solar PV array-based DC-DC converter with and without ZVS technique for small power applications. Compare with conventional topology, advantages of DC–DC converters with ZVS technique is more efficient. From the response study of ZETA and SEPIC DC-DC converter, rise in efficiency using ZVS technology are 1.97% and 7.23% respectively as compare to conventional converter in P & O MPPT control algorithm. Likewise, 4.14% and 0.41% rise in efficiency of ZVS based ZETA and SEPIC converter respectively in IC MPPT control algorithm. The system dynamic performances are also studied and found effective. The solar plates produce exact characteristic curves under P &

O and IC MPPT Control. By using the ZVS strategy, only the switching losses are negligible and the remaining losses are same as conventional topology. The key drawback of ZVS circuit is bulkier due to the using additional passive components. From the results study, the IC approach offers higher efficiency as compare to P & O approach. In this paper, the simulation and experimental study of only ZETA converter with IC MPPT is analysed under static and dynamic conditions. By employing ZETA converter in the solar connected system, it will regulate the output voltage coming from the solar panel based on various domestic applications such as solar charger for automotive batteries, lighting system, power supply etc.

Conflict of interest

No conflict of interest regarding this manuscript.

Appendix A

A1. System parameters for ZETA converter

$V_{PV} = 17$ V, $I_{PV} = 0.58$A, Linear load resistance $R = 60\Omega$, L1=L2 = 5.208 mH, $L1_{ESR}$=$L2_{ESR} = 83$ mΩ, C1 = 208.3μF, C2 = 21.69μF, $C1_{ESR}$ =$C2_{ESR} = 5$mΩ, MOSFET $R_{ON} = 1.2$mΩ, MOSFET $R_{in} = 0.01$ Ω, $L_r = 848.82\mu$H, $C_r = 190.985$ nF, Switching frequency (f_s) = 5 kHz.

A2. System parameters for SEPIC converter

$V_{PV} = 17$ V, $I_{PV} = 0.58$A, Linear load resistance $R = 60$ Ω, L1 = L2 = 4.309 mH, $L1_{ESR}$=$L2_{ESR} = 83$ mH, C1 = 315μF, C2 = 210μF, $C1_{ESR}$=$C2_{ESR} = 5$mΩ, MOSFET $R_{ON} = 1.2$mΩ, MOSFET $R_{in} = 0.01\Omega$, $L_r = 848.82\mu$H, $C_r = 190.985$ nF, switching frequency (f_s) = 5 kHz.

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